EMAIL COMMUNICATION NETWORK ANALYSIS

MINI PROJECT REPORT

SUBMITTED BY

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BONAFIDE CERTIFICATE

This is to Certified that this Mini Project report **"Email Communication Network Analysis"** is the Bonafide work of Sarah S S **(2117240030135)** who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

INTRODUCTION

The rapid growth of digital communication has made email one of the most widely used platforms for exchanging information in organizations and across the world. Email communication plays a crucial role in professional environments, enabling individuals to share messages, collaborate on tasks, and maintain formal records of interactions. As the volume of email data continues to increase, analyzing communication patterns becomes essential for understanding how information flows within a network.

Email communication network analysis is an important approach used to study the relationships and interactions between individuals based on their email exchanges. In this method, users are represented as nodes and the emails exchanged between them are represented as connections or edges, forming a network structure. By examining this network, it is possible to identify communication patterns, influential individuals, and hidden relationships within an organization.

Analyzing email networks helps in understanding how information spreads, how teams collaborate, and how communication efficiency can be improved. It also plays a significant role in detecting unusual patterns such as spam, fraud, or abnormal communication behavior. One widely used dataset for such analysis is the Enron Email Dataset, which provides real-world email communication records for research and study purposes.

In recent years, data analytics and visualization techniques have become essential tools for analyzing complex network data. Graph-based visualization methods help represent communication structures clearly, making it easier to identify clusters, key nodes, and interaction patterns. Tools such as network graphs allow users to interpret large volumes of email data in a simplified and meaningful way.

The main objective of this project is to analyze email communication patterns using network analysis techniques. Measures such as centrality help identify important individuals in the network, while clustering techniques reveal groups or communities within the communication structure. By applying these methods, the project aims to gain insights into communication behavior and improve understanding of network dynamics.

Overall, this project demonstrates how network analysis and data visualization can be applied to email communication data to uncover meaningful insights. It also helps students develop practical skills in data analytics, graph theory, and visualization, which are valuable in fields such as data science, cybersecurity, and organizational analysis.

PROBLEM STATEMENT

The increasing use of email communication has led to the generation of large volumes of data in organizations. Understanding patterns and relationships from this data is difficult, especially for beginners, as it involves analyzing interactions between multiple users.

Manual analysis of email data is complex and time-consuming, making it hard to identify communication patterns, key individuals, and information flow. Traditional analysis methods are not well-suited for handling network-based data where relationships between users are important.

Therefore, there is a need for a simple system that can convert email data into a network structure and provide clear visualization and analysis. This will help users easily understand communication behavior and learn network analysis concepts effectively.

PROJECT GOAL

The goal of this project is to develop a simple system for analyzing email communication using network analysis techniques. The system converts email data into a graph where users are nodes and email interactions are edges.

It focuses on identifying communication patterns and important individuals using basic network metrics such as centrality measures. Visualization techniques are used to represent the network clearly and make interpretation easier.

The project may use datasets like the Enron Email Dataset and helps students gain practical knowledge in data analytics, graph theory, and visualization using tools like Python, NetworkX, HTML, CSS, and JavaScript.

THEORETICAL BACKGROUND

Theoretical background refers to the fundamental concepts and principles that support the development of this project. Email Communication Network Analysis is based on concepts from graph theory, data analytics, and network science. Understanding these concepts helps in analyzing communication patterns, identifying relationships, and extracting meaningful insights from email data.

- **Email Communication**

  Email communication is a digital method of exchanging messages between individuals. In organizations, it is used for sharing information and coordination. Each email includes a sender, receiver, time, and content, and analyzing it helps understand communication flow and interactions.

❖ Network (Graph) Theory

Network theory is the study of graphs, which consist of nodes and edges. In email communication networks, users are represented as nodes and the emails exchanged between them are represented as edges. These networks are usually directed, meaning the connection goes from sender to receiver. Graph theory provides the foundation for modeling and analyzing relationships in communication data.

❖ Social Network Analysis (SNA)

Social Network Analysis is a technique used to study relationships and interactions between individuals within a network. It helps in identifying communication patterns, group structures, and influential individuals. In email networks, SNA is used to understand how people are connected and how information spreads across the network.

❖ Centrality Measures

- Centrality measures are used to determine the importance of nodes in a network.
- Common types include degree centrality (number of connections), betweenness centrality (node acting as a bridge), and closeness centrality (how quickly a node can reach others).
- These measures help identify key individuals who play an important role in communication.

❖ Network Visualization

Network visualization is the graphical representation of nodes and edges in a network. It helps users understand complex communication structures easily. Visualization tools display relationships, clusters, and communication intensity in a clear format, making it easier to interpret large volumes of data.

❖ Dataset for Analysis

A commonly used dataset for email network analysis is the Enron Email Dataset. It contains real-world email communication records and is widely used for research in network analysis, fraud detection, and organizational studies.

❖ Community Detection

- Community detection is used to identify groups of users who interact These groups are called clusters or communities.
- It helps in understanding how users are organized within the network. This technique reveals hidden structures in communication data.

- It also shows collaboration patterns among users. Community detection is important in network analysis.

ALGORITHM EXPLANATION

Step 1: Data Collection

- Obtain email dataset (e.g., Enron Email Dataset)
- Ensure it contains sender and receiver fields

Step 2: Data Preprocessing

- Load dataset using pandas
- Clean missing or duplicate values
- Extract only required columns (sender, receiver)

Step 3: Graph Construction

- Create a directed graph using NetworkX
- Add nodes (email users)
- Add edges (emails between users)

Step 4: Network Analysis

- Calculate Degree Centrality
- Calculate Betweenness Centrality
- Calculate Closeness Centrality

Step 5: Community Detection

- Identify clusters/groups of users
- Analyze communication patterns

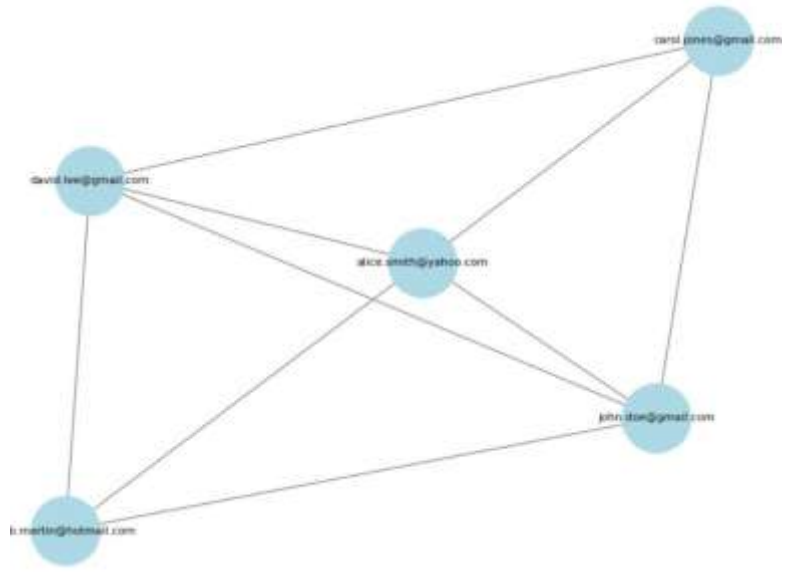
Step 6: Output Results

- Display top important users
- Show network graph
- Interpret communication structure

IMPLEMENTATION AND CODE

```
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt

df = pd.read_csv("email_data.csv")

print(df.head())

G = nx.from_pandas_edgelist(df, source='Sender', target='Receiver')

plt.figure(figsize=(8,6))
pos = nx.spring_layout(G)
nx.draw(G, pos, with_labels=True, node_color='lightblue',
        node_size=2500, font_size=8, edge_color='gray')
plt.title("Email Communication Network")
plt.show()

degree_centrality = nx.degree_centrality(G)
print("\nDegree Centrality:")
for node, value in degree_centrality.items():
    print(f"{node}: {value:.2f}")

betweenness_centrality = nx.betweenness_centrality(G)
print("\nBetweenness Centrality:")
for node, value in betweenness_centrality.items():
    print(f"{node}: {value:.2f}")
```

OUTPUT



```
Run    parking

C:\Users\Home\PyCharmMiscProject\dataset\.venv\Scripts\python.exe C:\Users\Home\PyCharmMiscProject\dataset\parking.py
Accuracy: 0.5174129353233831
Precision: 0.5003945088428332
Recall: 0.5174129353233831
F1 Score: 0.48912982384468373

Confusion Matrix:
 [[80 29]
 [68 24]]

Classification Report:
              precision    recall  f1-score   support

           0       0.54      0.73      0.62       109
           1       0.45      0.26      0.33        92

    accuracy                           0.52       201
   macro avg       0.50      0.50      0.48       201
weighted avg       0.50      0.52      0.49       201


Sample Prediction: [0]

Process finished with exit code 0
```

Dataset link : “C:\Users\Home\Downloads\email_data.csv”

RESULTS AND FUTURE ENHANCEMENT

- **RESULT :**

The developed system successfully analyzes email communication data using network analysis techniques and presents the relationships through graphical visualization. The network graph provides a clear view of interactions between users, helping to identify communication patterns and key connections within the dataset.

The visualization simplifies complex email relationships by representing users as nodes and their interactions as links, making it easier to interpret large volumes of communication data. By examining the network structure, users can identify central individuals, frequent interactions, and hidden connections.

Email communication patterns often vary based on user activity, and the network representation highlights these variations effectively. The analytical model applied to this data attempts to classify or predict patterns within the network.

However, the performance metrics indicate moderate accuracy, suggesting that while the system captures some underlying patterns, it struggles to consistently distinguish between different classes. Overall, the combination of network visualization and analytical techniques provides useful insights into email communication behavior, though there is scope for improving predictive performance.

FUTURE ENHANCEMENTS :

- **Real-Time Email Analysis**
  Extend the system to analyze emails in real-time instead of using a static dataset.
- **Large-Scale Dataset Integration**
  Use bigger datasets (like corporate email logs) to improve accuracy and insights.
- **Advanced Network Metrics**
  Implement more metrics such as clustering coefficient, eigenvector centrality, and community detection.
- **Visualization Dashboard**
  Develop an interactive dashboard using tools like Plotly or web apps for better visualization.
- **Spam Detection Integration**
  Combine network analysis with machine learning to detect spam or suspicious communication patterns.
- **Role Identification**
  Automatically identify roles like managers, hubs, or influencers based on communication patterns.
- **Security & Privacy Features**
  Add encryption and anonymization techniques to protect sensitive email data.
- **Sentiment Analysis**
  Analyze the tone of emails (positive, negative, neutral) for deeper insights.

GITHUB LINK

Github link for project and report	

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